

# Pranshu Gupta

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## PROFESSIONAL EXPERIENCE

- **Microsoft AI** / *Member of Technical Staff* / Redmond, WA Aug 2025 – Present
  - Engineered the inference integration for **GPT-5.1 instant and reasoning models** on Microsoft Copilot — prompt templates, constrained sampling, chain-of-thought handling, and safety orchestration — delivering a **30% improvement in personality evals** for the reasoning path.
  - Eliminated non-essential classifiers and summarizers on the instant path, reducing GPU footprint by **1,320 H100 GPUs**.
  - Discovered and fixed a bug in the OpenAI Chat Completions API causing function-definition parameter descriptions to be duplicated, saving **6K tokens per inference call** and reducing **TTFT by 30ms**.
  - Designed **multi-region routing architecture** for the M365 LLM inference API: health-signal-driven failover with ordered regional fallback, preserving GPU session affinity across completions API server failures.
  - Designed and shipped **Adaptive Thinking**, a dynamic system that adjusts the thinking effort per message based on intent/capability classification to control reasoning token usage.
  - Defined and instrumented LLM API **traffic classification** dimensions (product, client app, inference intent, license tier) across MAI and M365 Copilot; built Grafana dashboards to surface per-dimension traffic insights.
  - Architected and shipped an autonomous **code simplification agent** that scans recent commits, scopes safe refactors via durable git baselines, and manages the full PR lifecycle through automated validation.
- **Microsoft Azure AI** / *Senior Software Engineer* / Redmond, WA Jan 2025 – Aug 2025
  - Designed a distributed **token-based rate limiter** using Tiktoken for precise token counting, replacing a length-based estimator with **219% average error**; implemented pub-sub quota state synchronization across instances.
  - Contributed to an intelligent **model router** for per-request model selection based on cost/complexity signals, reducing per-inference cost by **55%**.
- **Microsoft Azure Core** / *Senior Software Engineer* / Redmond, WA Mar 2024 – Jan 2025
  - Designed and shipped a token-based job throttling system with multi-bucket quota enforcement and retry-storm protection, preventing partition-level noisy-neighbor throttling in background-job infrastructure of Azure Control Plane.
  - Single-handedly designed and shipped async operation URI signing, adopted by **242 Azure service teams** and enforced for **220+ resource providers**.
- **Microsoft Azure Core** / *Software Engineer II* / Redmond, WA Jun 2020 – Mar 2024
  - Designed and shipped async operation callbacks replacing polling, reducing ARM-to-resource-provider traffic by up to **94%** (130M fewer requests/day) and improving p95 latency by **12 seconds** for Azure Container Instances.
  - Extended and maintained **Azure DevOps CI/CD pipelines** for ARM control-plane services with staged multi-region rollouts, 24-hour regional bake times, functional test gates, and rollback-via-redeploy across global Azure regions.
  - Led **Azure region teardown** end-to-end: built workflows to halt new resource creation, resolve subscription/resource-group hierarchies, and migrate live resources to healthy regions with zero data loss.
- **Microsoft Core Services** / *Software Engineer* / Hyderabad, India Jun 2017 – Aug 2019
  - Modernized a data enrichment platform to Azure Function Apps (**\$4K/month** savings); integrated Cognitive Services for data standardization (**150K leads/month**) and built a bulk-processing pipeline with **40–50x throughput** improvement.

## TECHNICAL SKILLS

- **AI/ML & Inference:** LLM inference/serving, prompt engineering, model evaluations, safety orchestration, RAG
- **Languages & Systems:** Python, .NET, C++, KQL, JavaScript, HTML, CSS, Kubernetes, Redis, Grafana
- **Cloud & Infrastructure:** Azure (Cosmos DB, ARM Templates, Service Bus, DevOps), AWS (Bedrock, S3, ECR)

## SELECTED PROJECTS

- **AI Lab Assistant** (TurnTheBus nonprofit, 2025–Present): Architected and built end-to-end on **AWS** using **RAG** over physics lab manuals to guide high school students through virtual physics experiments.

## PUBLICATIONS & WRITING

- *Runtime Optimizations for LLMs* (2026). Production notes on inference optimizations for latency/cost-efficient LLM serving.
- *The Observer Effect on Social Media* — ACM CHI (2024). Causal inference and time series study on longitudinal Facebook data from 300+ participants to quantify how monitoring awareness alters online behavior.
- *Mental Health Impact of Active-Shooter Drills* — Nature (2021). Applied ML classifiers and interrupted time series analysis to 54M social media posts across 114 schools to measure psychological and civic impact.

## EDUCATION

- **Georgia Institute of Technology** / *M.S. Computer Science (Machine Learning)* / Atlanta, GA May 2020
- **Indian Institute of Technology, Kanpur** / *B.Tech, C.S. and Eng.* / Kanpur, India May 2017